

Bitwise linear trail through **ARADI** found by SAT

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1 ARADI

Maximal linear correlation: 2^{-4}

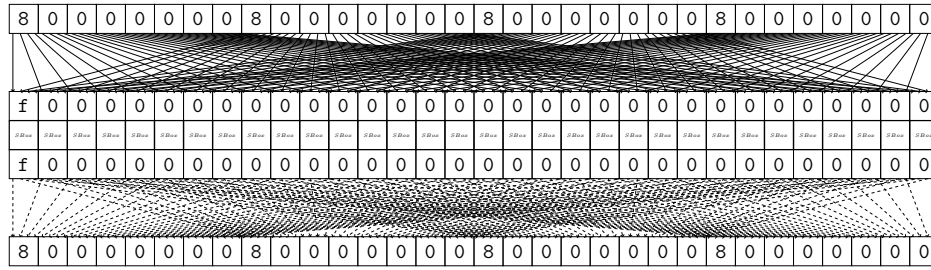
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
ARADI - round0																															
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
ARADI - round1																															
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2
PostBK																															
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2

2 ARADI-round0

Maximal linear correlation: $2^{-1.0}$

8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
BN																													
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
SBias Layer																													
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	
Linear Layer0																													
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	2

3 SBoxLayer

Maximal linear correlation: $2^{-1.0}$ 

4 LinearLayer0

Maximal linear correlation: $2^{-0.0}$

8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0
L0								L0								L0								L0															
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2

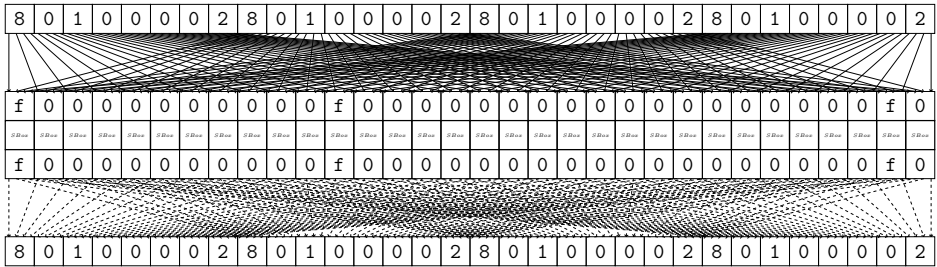
5 ARADI-round1

Maximal linear correlation: $2^{-3.0}$

8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
RK																																							
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
SBoxLayer																																							
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
LinearLayer1																																							
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2

6 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



7 LinearLayer1

Maximal linear correlation: $2^{-0.0}$

8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2	8	0	1	0	0	0	0	2
S1								S1								S1								S1															
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2
8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2	8	5	3	0	0	2	9	2